From Orders to Opportunities: Strategic Insights for Miami-Dade County Suppliers

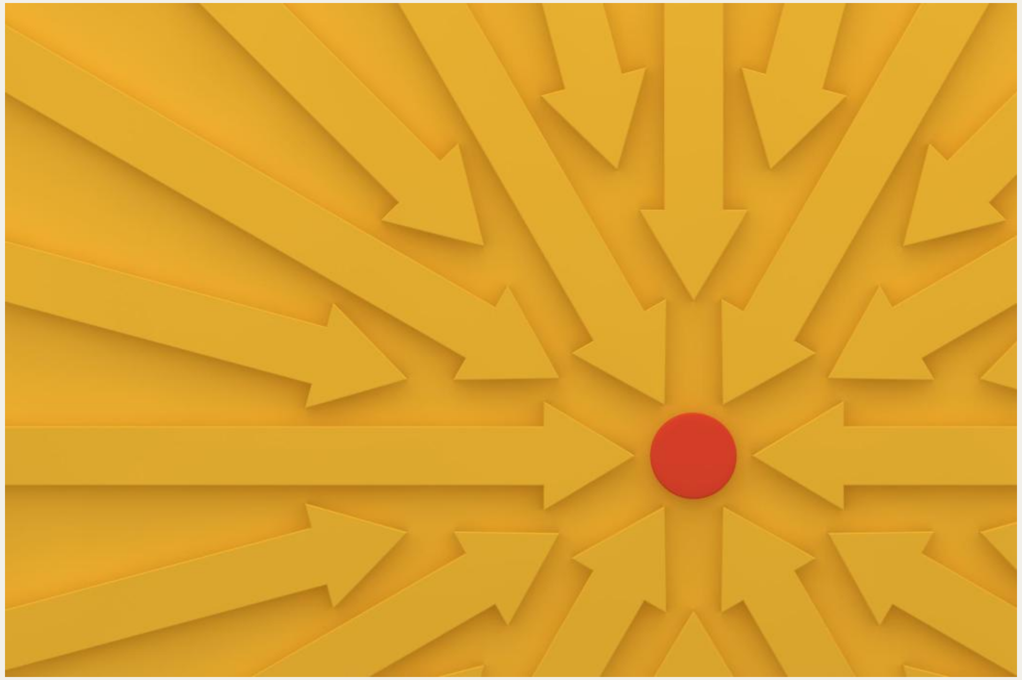
Presented by David Perez
Presented on April 30, 2025

THE AGENDA

- Introducing the objective
- The methodology of attaining the objective
- Analysis and results
- Case study
- Recommendations
- Conclusion
- Appendix

INTRODUCTION

THE OBJECTIVE



Identifying opportunities for Miami-Dade County to utilize its significant purchasing power to negotiate more favorable agreements with its suppliers.

PURCHASING POWER

In 2024, Miami-Dade County allocated more than $4B in expenses. Given such significant purchasing power, there must be opportunities to strategically leverage this financial capacity.

2024 PURCHASE ORDER SUMMARY

- ✓ 43,058 total transactions
- ✓ 31,069 unique items
- ✓ 1,383 various suppliers
- ✓ Total expenditure of $4,288,751,653.82

METHODOLOGY

IDENTIFYING OPPORTUNITY

We will be using the Miami-Dade County 2024 Purchase Order dataset to conduct a “Reverse” RFM Analysis and Monte Carlo Analysis. We will be also demonstrating a case study to show this methodology in action.

WHAT IS AN RFM ANALYSIS?

An RFM Analysis evaluates transactions based on three dimensions: Recency (how recently a transaction occurred), Frequency (how often transactions take place), and Monetary Value (the total monetary number of transactions).

WHAT IS A MONTE CARLO ANALYSIS?

A Mote Carlo Analysis is a multiple random simulation technique employed to assess the likelihood of various potential outcomes that assist in risk evaluation and informed decision making.

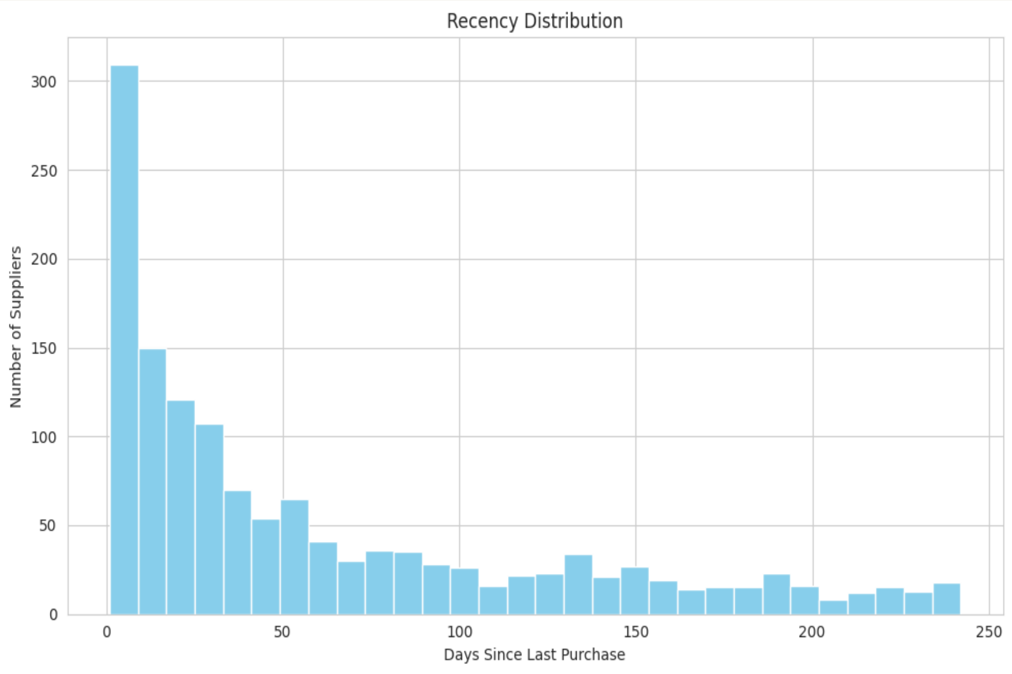
USING RFM AND MONTE CARLO

We employed RFM analysis to assess the recency, frequency, and monetary value of purchases made by Miami-Dade County for each supplier.

The Monte Carlo Analysis provides an estimation of Miami-Dade County's projected value to suppliers by forecasting its spending in 2025.

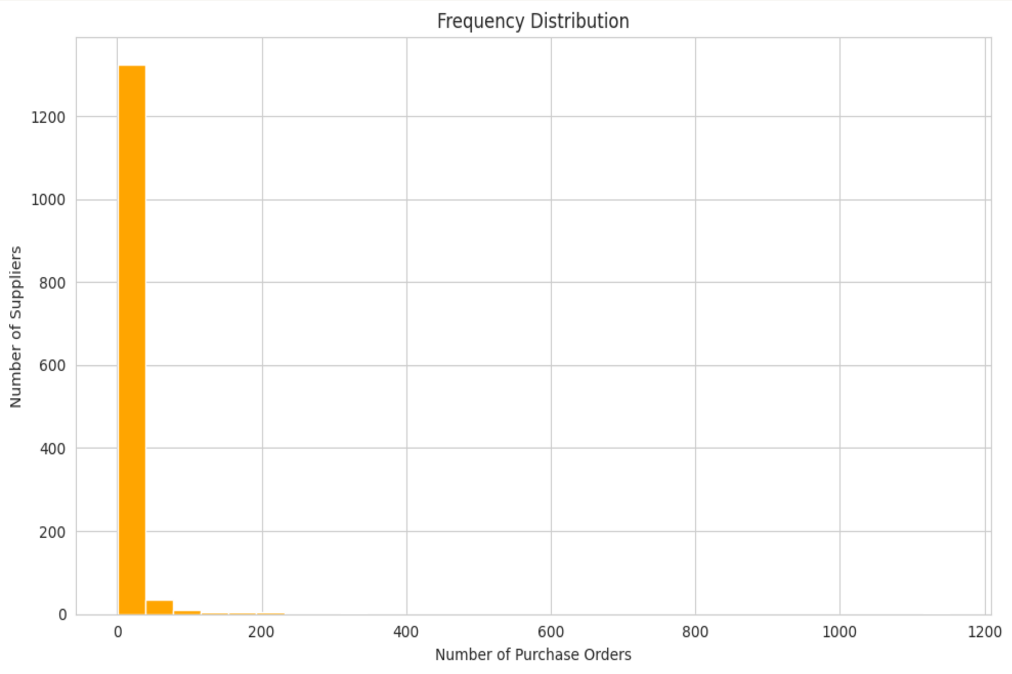
ANALYSIS

THE RECENCY



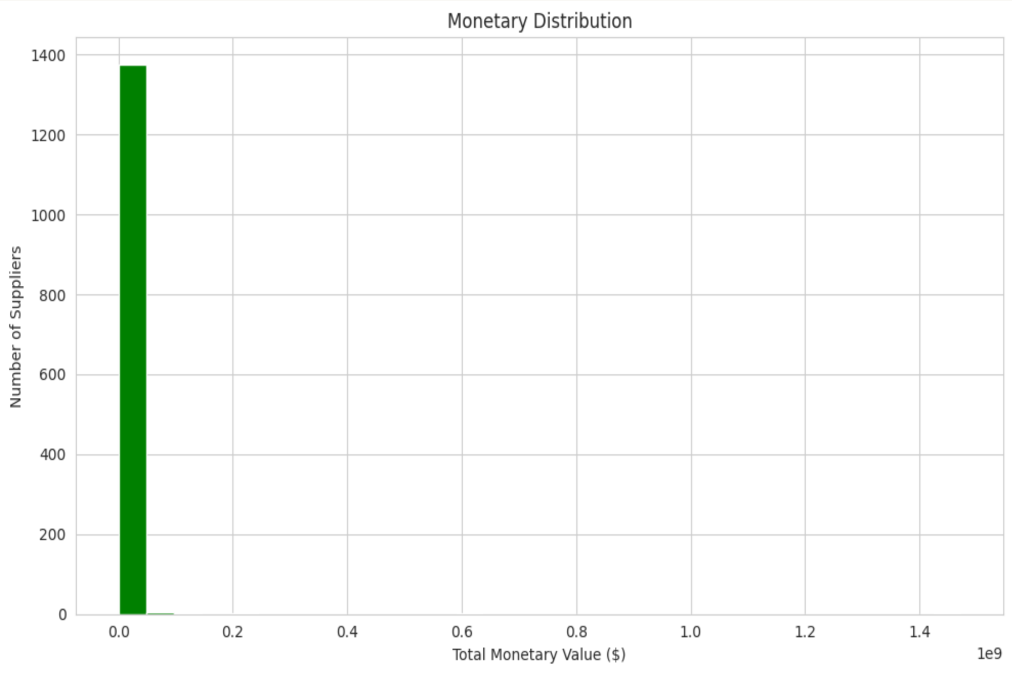
The recency distribution indicates a downward trend, suggesting that a significant quantity of items have been acquired in the recently. This implies that there may be some negotiating power regarding recency when engaging with suppliers.

THE FREQUENCY



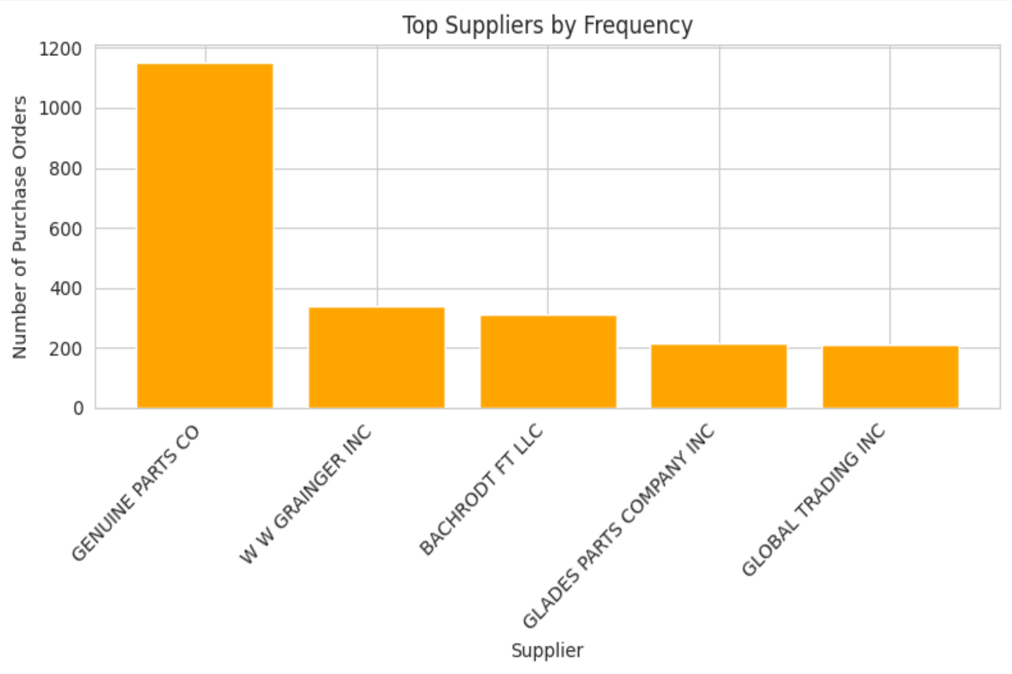
The frequency distribution shows us that a significant volume of purchases is directed towards the same suppliers. This indicates potential leverage in frequency when engaging in negotiations with suppliers.

THE MONETARY



The monetary distribution shows us that most suppliers are being compensated at roughly equal levels. This suggests a potential avenue to enhance cost efficiency during negotiations with these suppliers.

SUPPLIER EVALUATION



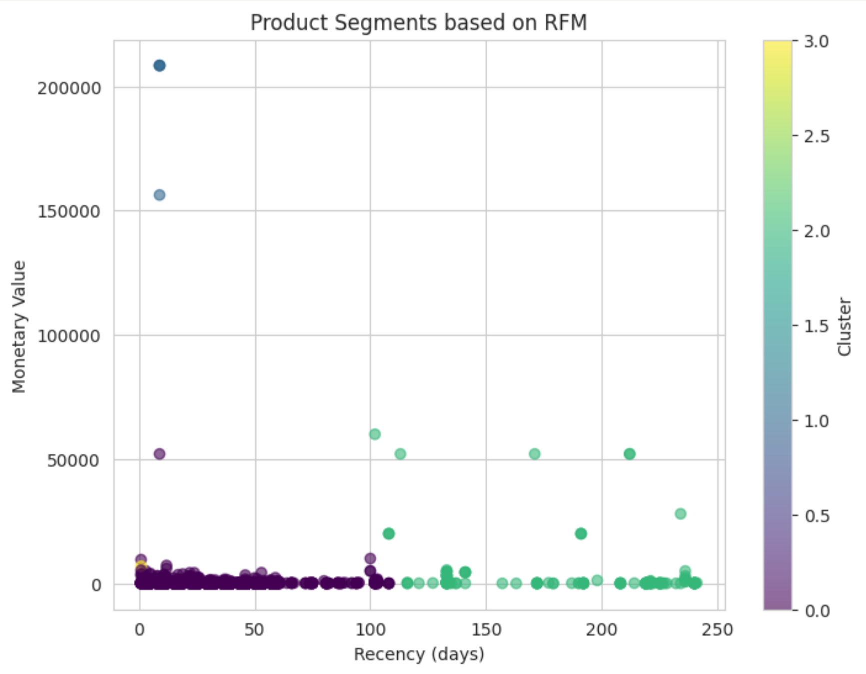
We identified the primary suppliers from whom Miami-Dade County made the most frequent purchases, those with the highest monetary value, and the suppliers with the most recent transactions. To illustrate how this data can be beneficial for business negotiations, we chose to evaluate Genuine Parts Co., the supplier with the highest purchase frequency, as a case study.

CASE STUDY

GENUINE PARTS CO

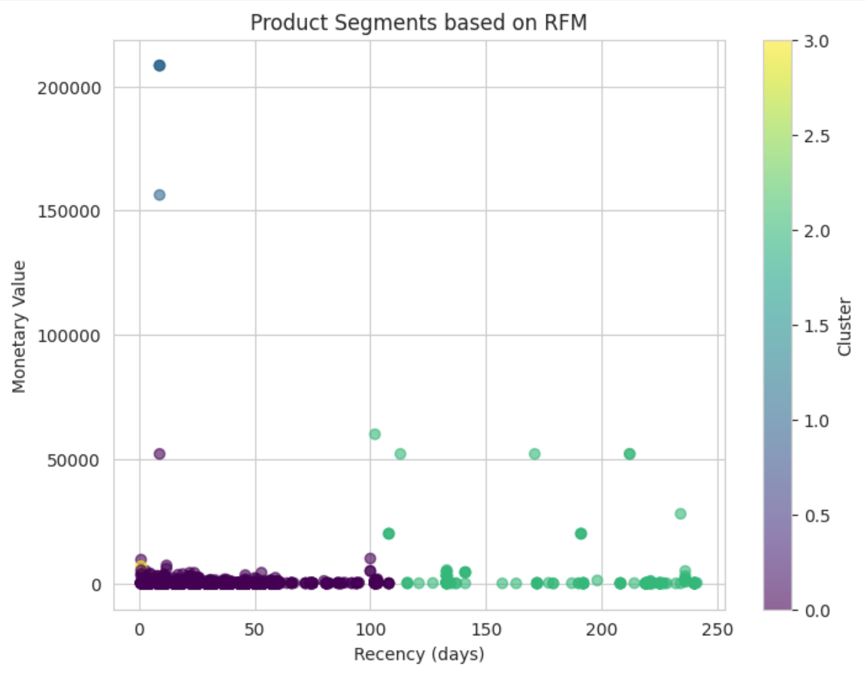
Miami-Dade County expended $876,747.18, acquiring 2,276 distinct items from Genuine Parts Co in 2024.

PRODUCT SEGMENTATION



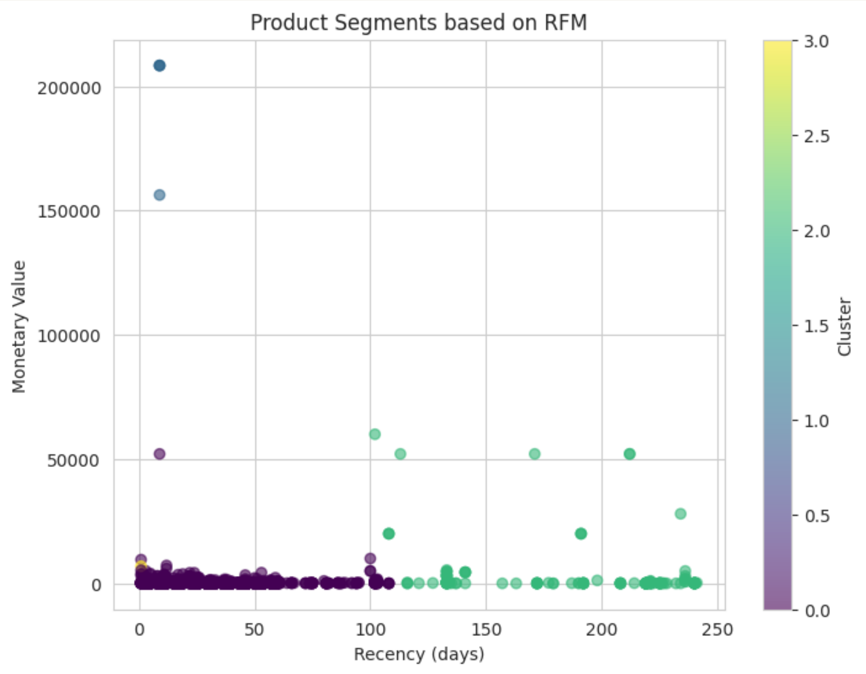
We employed a clustering algorithm based on our RFM Analysis to categorize products from Genuine Parts Co into distinct groups.

SEGMENTATED CLUSTER 0



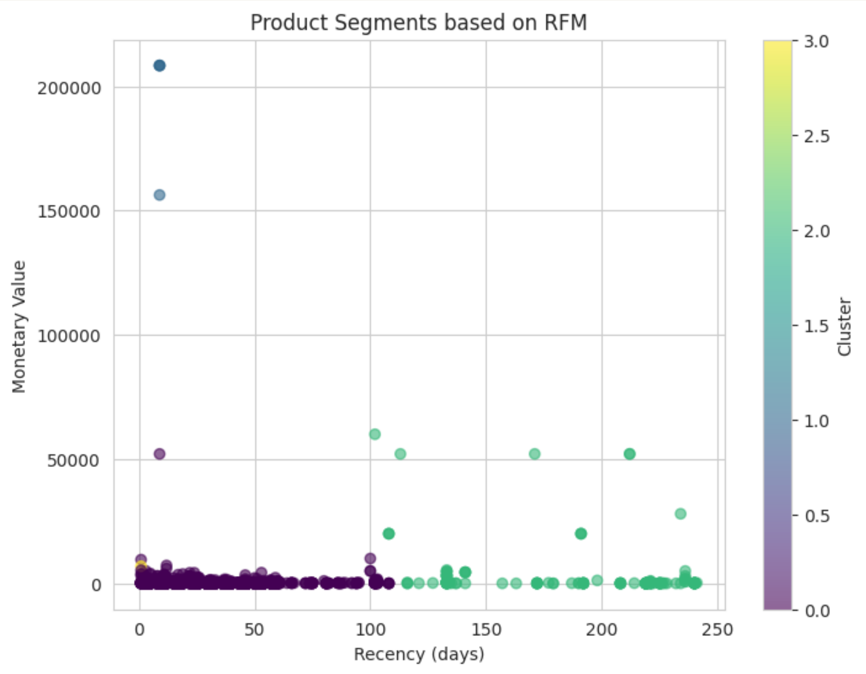
The Cluster 0 group (Purple) contains a total of 2,132 items, characterized by a low monetary value per item overall, yet these items are frequently and recently purchased.

SEGMENTATED CLUSTER 1



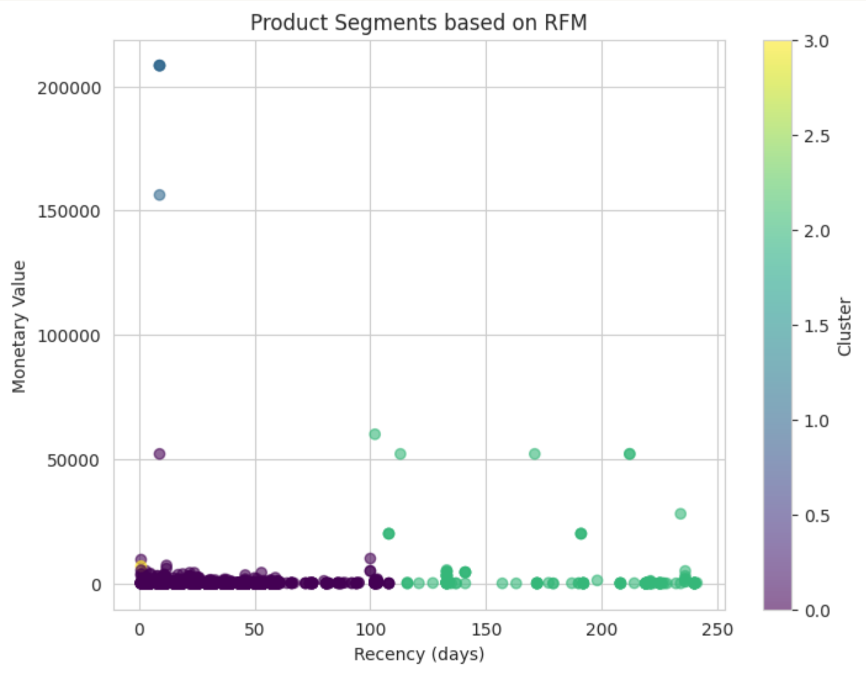
The Cluster 1 group (Blue) includes 4 items, representing unusually large purchases that are not made very often.

SEGMENTATED CLUSTER 2



The Cluster 2 group (Green) has 148 items, which are infrequently purchased and have the second lowest monetary value among the four clusters.

SEGMENTATED CLUSTER 3

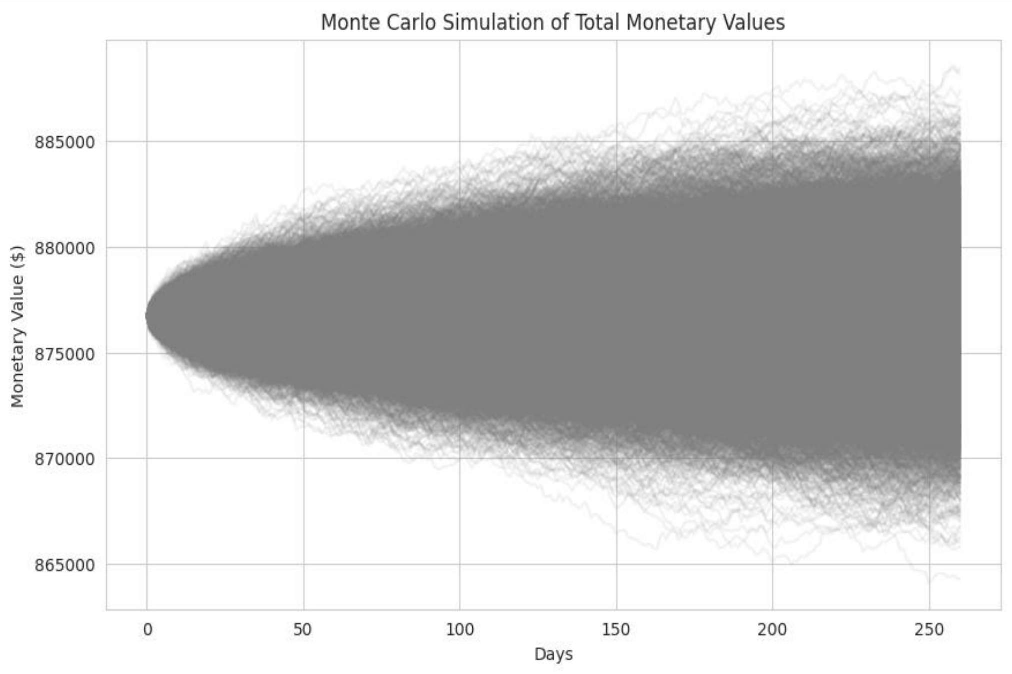


The Cluster 3 group (Yellow) consists of a single core item, which has been purchased 66 times and was acquired recently.

SEGMENTATION CONCLUSION

Clusters 0 and 3 demonstrate the highest potential, with Cluster 0 exhibiting a total monetary value of over $572K and Cluster 3 showing a total of almost $7k. The items within these clusters are frequently purchased and been acquired recently, making them ideal candidates for negotiation discussions with the supplier.

MONTE CARLO ANALYSIS



After conducting 10,000 simulations with a 95% confidence interval over a business year, Miami-Dade County's forecasted value to GPC in 2025 is $875K with any variation falling between $872K and $877K. We will use this data to highlight potential losses if Miami-Dade County decides to shift their business elsewhere.

RECOMMENDATIONS

DISCOUNTED RATES

- ✓ Miami-Dade County is eligible to seek an overall discount on the anticipated expenditure of $875K with Genuine Parts Co in 2025.

- ✓ Miami-Dade County has the option to request that discounts be structured in tiers. For example, a 10% discount will be applied upon exceeding the $100k mark, followed by a 15% discount once the $300k threshold is reached, and so on.

SEGMENTED DISCOUNTS

- ✓ Cluster 0 holds a total monetary value of over $572K. Given its substantial value, it may be feasible to apply an overall discount to all items within this cluster.
- ✓ Cluster 3 has a monetary value of almost $7K, attributed to a single core item that was acquired 66 times in 2024. This item could be a candidate for a discount, considering its frequent purchases.

NEGOTIATE FOR SPECIFIC ITEMS

Negotiate for items that exhibit a strong combination of frequency and recency. For example, if an item’s frequency is 12 or greater, and has a recency threshold of 30 days or fewer, we can infer that these items are likely purchased monthly. With this data, Miami-Dade County can advocate for monthly discounts on these products.

CONCLUSION

THE SUMMARY

This comprehensive analysis can be applied to any supplier within the dataset and serves as a valuable negotiating tool for securing improved pricing. Achieving better pricing could lead to Miami-Dade County saving potentially hundreds of thousands of dollars, or even more based on negotiations, which can then be redirected towards other county initiatives. Furthermore, this data may provide leverage in discussions with rival suppliers to obtain more favorable pricing if current suppliers are unwilling to negotiate.

APPENDIX

FURTHER ANALYSIS

- ✓ Access to completed datasets from prior years can enhance granularity, enabling more precise RFM and Monte Carlo predictions.
- ✓ Competitor price sheets would also be beneficial during negotiations.
- ✓ Any contracts or agreements currently in use with suppliers.
- ✓ Incorporating categorical data could further bolster the negotiation strategy.
- ✓ Collaborating with specialists in each supplier domain or developing a detailed guide on supplier business relations for various sectors would also prove advantageous.

LINKS

Application Link: https://miami-dade-po-analysis.netlify.app/
Github Link: https://github.com/Perez786/DataMining_FinalProject
Dataset Link: https://gis-mdc.opendata.arcgis.com/datasets/673f9be756ce4d2a95039bf16d3a3f7d_0/explore